

Alkali diffusivity in alkaline earth sodium boroaluminosilicate glasses



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ARTICLE INFO

Article history:

Received 8 April 2014

Received in revised form 23 May 2014

Accepted 23 May 2014

Available online 7 June 2014

Keywords:

Diffusion

Amorphous solids

Ion exchange

Boroaluminosilicate

ABSTRACT

Understanding the fundamentals of alkali diffusion in boroaluminosilicate glasses is of critical importance for advanced glass applications, where the alkali diffusivity should be either enhanced or reduced. Here, we investigate the sodium–potassium interdiffusivity in thirty-six alkaline earth sodium boroaluminosilicate glasses with varying Si/Al ratio and type of alkaline earth cation by ion exchange experiments. The alkali mobility is maximized when Na⁺ acts as a charge compensator to Al³⁺ or B³⁺ in the glass rather than as a creator of non-bridging oxygen. In the peralkaline regime, the larger alkaline earth cations significantly slow down the motion of alkali ions, while this effect is much smaller in the peraluminous regime. Generally, there is a larger effect of increasing temperature on diffusivity for the larger alkaline earth cations than for the smaller ones.

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1. Introduction

The transport properties of oxide glasses have been of primary interest of both the fundamental research as well as the industrial research which focuses on practical applications in different technologies [1,2]. Boroaluminosilicate glasses are particularly widely used in the industry, since the mixing of the three network former cations (Si⁴⁺, Al³⁺, B³⁺) allows for delicate tuning of the glass properties [3]. Understanding the composition dependence of alkali ion transport in boroaluminosilicate glasses is of critical importance not only for chemically strengthened ion exchanged glasses [4–14], but also for boroaluminosilicate glasses as substrates for LCD/LED displays [15,16] and photovoltaics [17,18]. In the substrate applications, the diffusivity of the alkali ions should be kept as low as possible to avoid interference with the devices grown on these substrates. Hence, it is important to understand the composition and temperature dependence of alkali diffusion in these glasses in order to compositionally design new glasses with either enhanced or reduced alkali diffusion depending on the application.

Understanding the alkali diffusion characteristics is also of fundamental interest, e.g., for improvement of alkali mobility models and structural understanding of glasses and glassy dynamics [19,20]. Such understanding requires that the influence of chemical composition on network structure is well understood [21]. In this work, we study the diffusion kinetics of the K⁺-for-Na⁺ interdiffusion process, which has previously been investigated for various sodium silicate [22], sodium borosilicate [22], sodium aluminosilicate [23], sodium alkaline earth silicate [24], and sodium boroaluminosilicate [5] glass systems. For example, it has been shown that in sodium boroaluminosilicate glasses,

sodium is more mobile when it acts as a charge compensator to stabilize network formers than when it acts as a creator of non-bridging oxygens on tetrahedrally-coordinated silicon and trigonal boron [5]. However, many industrial compositions also contain alkaline earth cations, which are competing with the sodium cations for the role as charge compensator of tetrahedral aluminum and boron groups. In this study, we cast light on the impacts of network former speciation and modifier field strength on the alkali diffusivity in alkaline earth sodium boroaluminosilicate glasses over a wide compositional regime.

2. Experimental

We prepared a total of thirty-six alkaline earth sodium boroaluminosilicate glasses with either magnesium, calcium, strontium, or barium (in mol%): (76 – x)SiO₂–xAl₂O₃–6B₂O₃–12Na₂O–6RO with x = 0, 3, 6, 9, 12, 15, 18, 21, and 24 for R = Mg, Ca, Sr, and Ba. We also included 0.15 mol% SnO₂ as fining agent in all compositions. The preparation has been described in detail elsewhere [25,26]. All glasses were annealed for 2 h at their respective annealing points followed by slow (1 K/min) cooling. The analyzed chemical compositions of the final glasses were found to be consistent with the nominal ones within ± 0.5 mol%.

Ion exchange experiments were conducted in order to obtain the effective interdiffusion coefficient $\bar{D}_{\text{Na-K}}$ between Na⁺ and K⁺. This was done by immersing polished 25 × 25 × 1 mm³ samples in a molten salt bath of technical grade KNO₃ at 370, 390, 410, 430, and 450 °C for 8 and 16 h. Consequently, the large alkali ions (K⁺) from the molten salt bath exchange with the smaller alkali ions (Na⁺) in the host glass. This process results in the formation of a surface compressive stress. The penetration depth of the potassium ions was measured using an FSM-6000 instrument (Frontier Semiconductor). The K⁺-for-Na⁺

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interdiffusion gives the glass surface a higher refractive index than the interior, i.e., the surface can act as a waveguide. This is utilized in the FSM instrument to measure the saturation depth of the refractive index profile, which corresponds to the diffusion depth of potassium [27]. A total of eight FSM measurements were performed on each sample (using four 90° rotations per face).

3. Results and discussion

In the classical description of mutual diffusivity during an interdiffusion process, the interdiffusion coefficient D_{Na-K} is given by the Nernst–Planck equation,

$$D_{Na-K} = \frac{D_{Na}D_K}{D_{Na}N_{Na} + D_KN_K}, \quad (1)$$

where N_i and D_i are the fractional concentration and self-diffusion coefficient of alkali ion i , respectively. This equation is only valid under the assumption of thermodynamic ideality, i.e., $\partial \ln a_i / \partial \ln c_i = 1$, where a_i and c_i are the thermodynamic activity and concentration of alkali ion i , respectively. It is necessary to define the interdiffusion coefficient since the larger potassium ions diffuse at a slower rate than the smaller sodium ions, i.e., D_{Na-K} is different from the sodium self-diffusion coefficient (unless $N_{Na} = 0$). Furthermore, D_{Na-K} varies with local composition, temperature, and time [28]. We therefore define an average or effective interdiffusion coefficient ($\bar{D}_{Na-K}$) as a constant for a given diffusion profile. $\bar{D}_{Na-K}$ is calculated based on the potassium diffusion depth combined with the interdiffusion time and temperature. The optical measurements provide the diffusion depth of the potassium ions as a result of the K^+ -for- Na^+ interdiffusion process. This depth is calculated from the number of fringes and width of the fringe pattern based on the data of TM mode of light propagation. An example of an optical fringe pattern is shown in Fig. 1.

The determined composition dependence of $\bar{D}_{Na-K}$ at 410 °C is illustrated in Fig. 2. The composition dependence at the other investigated temperatures is essentially the same, despite the differences in activation energy discussed below. The data in Fig. 2 are plotted against the value of $[Al_2O_3] - [Na_2O]$, i.e., the excess concentration of Al^{3+} not being charge-balanced in tetrahedral configuration by Na^+ . For any given value of $[Al_2O_3] - [Na_2O]$, the diffusivity increases with decreasing size of the alkaline earth cation. The presence of relatively immobile alkaline earth cation lowers the alkali diffusivity and this effect is larger as the size of the divalent cation increases. As Al_2O_3 is substituted for SiO_2 and the network connectivity increases, the diffusivity exhibits a maximum value. The position of this maximum varies with the type of

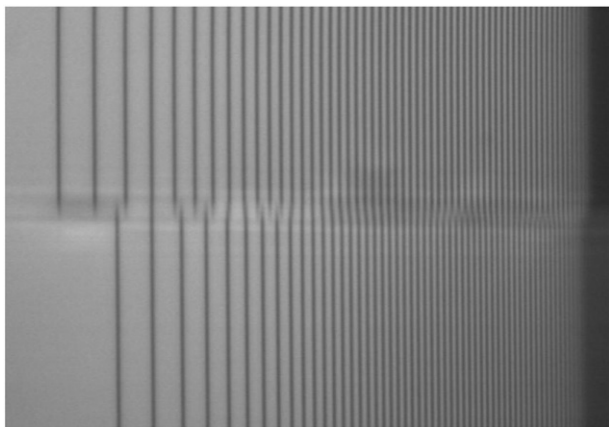


Fig. 1. Example of optical fringe pattern for a glass that has been immersed in molten KNO_3 . The upper set of fringes in the image corresponds to the TM mode of light propagation, while the lower set of fringes corresponds to the TE mode. The diffusion depth is calculated from the number of fringes and width of the fringe pattern based on the TM mode data.

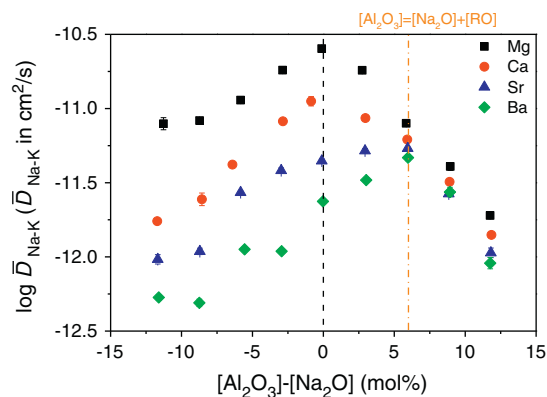


Fig. 2. Composition dependence of isothermal K^+ -for- Na^+ effective interdiffusion coefficient ($\bar{D}_{Na-K}$) as determined by interdiffusion experiments at 410 °C. The error bars of $\bar{D}_{Na-K}$ correspond to the standard deviation of the eight measurements of the potassium diffusion depth for the same sample.

alkaline earth cation, as the maximum diffusivity is found at a higher Al_2O_3 content with increasing alkaline earth size (Fig. 2). We find that for the two glasses with the highest field strength cations (F), i.e., Mg^{2+} and Ca^{2+} , the maximum value of the interdiffusion coefficient is around the $[Al_2O_3] = [Na_2O]$ compositional point. The other two glasses with Sr^{2+} and Ba^{2+} , which have smaller field strength values, exhibit the maximum in interdiffusion coefficient at the composition corresponding to $[Al_2O_3] = [Na_2O] + [RO]$. For $[Al_2O_3] - [Na_2O] - [RO] > 0$, the effect of the type of alkaline earth cation on the alkali diffusivity is much smaller.

For the low- Al_2O_3 regime, some of the sodium ions are used to create non-bridging oxygens (NBOs) with Si–O or B–O. These sodium ions are found to be less mobile, e.g., Terai [29] reported that sodium diffusivity achieves its maximum value when AlO_4 tetrahedra are completely charge-balanced by Na^+ and no NBOs exist. This agrees with the results shown in Fig. 2 for the Mg- and Ca-containing glasses. For $[Al_2O_3] - [Na_2O] > 0$, the charge compensation of Al^{IV} must be performed by the divalent alkaline earth cations, which are not as effective for charge compensation as the monovalent sodium ions due to the difference in field strength [30]. This could explain the difference in the position of the maximum point in Fig. 2. These changes in glass network structure with composition are also reflected in the composition dependence of various physical properties [25,26]. For example, there is an approximate linear decrease of diffusivity with increasing refractive index independent of the type of alkaline earth cation (Fig. 3), since higher

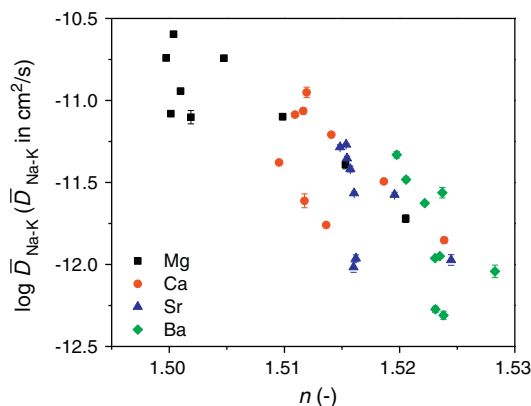


Fig. 3. Effective interdiffusion coefficient of K^+ -for- Na^+ ($\bar{D}_{Na-K}$) at 410 °C plotted as a function of the refractive index (n) at 589 nm of the sodium borosilicate glasses with magnesium, calcium, strontium, or barium. The uncertainty of n is ± 0.0003 . The values of n are taken from Ref. [26].

refractive index reflects tighter network packing and thus less space for ionic transport.

For all of the studied compositions, we have analyzed the temperature dependence of the sodium–potassium interdiffusivity. In the studied temperature range (370–450 °C), we have found that an Arrhenius equation could be used to fit the data and thus describe the temperature dependence of diffusivity,

$$\bar{D}_{\text{Na-K}} = \bar{D}_{\text{Na-K}}^0 \cdot \exp\left(\frac{-E_a}{RT}\right). \quad (2)$$

Here, E_a and $\bar{D}_{\text{Na-K}}^0$ are the activation energy and pre-exponential factor, respectively. Good agreement with the Arrhenius equation is found for the studied temperature range, as illustrated in Fig. 4 for the glasses with $x = 15$. The composition dependence of the fitted values of E_a is shown in Fig. 5a. There is no apparent trend in E_a with the SiO_2 -to- Al_2O_3 ratio. However, for a fixed alkaline earth cation, the average value of E_a for different $[\text{SiO}_2]/[\text{Al}_2\text{O}_3]$ ratios decreases with increasing field strength (i.e., decreasing size) of the alkaline earth cation (Fig. 5b). According to the modified random network (MRN) model for glasses proposed by Greaves [32], the network modifying oxides form interconnected channels (i.e., a percolative network) at sufficiently high concentration. The alkali cations should diffuse fastest through the channels when these are not “blocked” by the relatively immobile alkaline earth cations. The activation energy of diffusion in glasses is the sum of an electrostatic term due to the Coulomb interaction between the cation and the NBO plus an elastic part to open up doorways into neighboring sites [33], i.e., when the latter term governs the activation energy, E_a of alkali diffusivity will be smaller for smaller alkaline earth cation.

Finally, an interesting observation is that there is an apparent grouping of the four series of glasses. The two glass series with Mg and Ca exhibit maximum value of $\bar{D}_{\text{Na-K}}$ around the same composition (Fig. 2) and also have similar values of E_a (Fig. 5). This is also the case for the two glass series with Sr and Ba. A similar grouping of results has previously been reported for the composition dependence of coefficient of thermal expansion [26]. However, this apparent grouping cannot simply be explained by differences in field strength among the alkaline earth cations, since the field strength decreases smoothly from Mg to Ba. Nonetheless, it seems likely that it is related with the ability of the alkaline earth cations to charge-stabilize tetrahedral Al^{3+} and B^{3+} . If Mg and Ca are not as effective as Sr and Ba in stabilizing tetrahedral boron, the structure should be kept more open until $[\text{Al}_2\text{O}_3] = [\text{Na}_2\text{O} + \text{RO}]$ in the Sr- and Ba-containing glasses. Consequently, relatively fast alkali diffusivity is promoted.

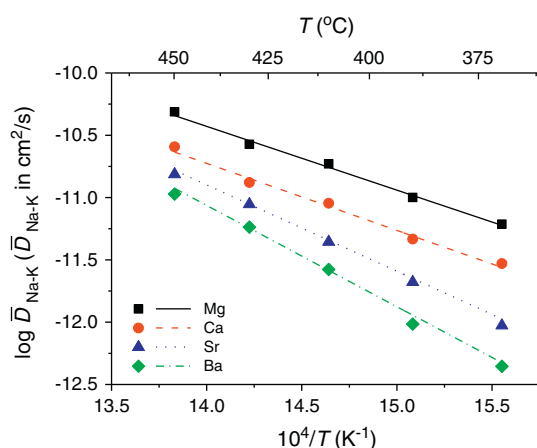


Fig. 4. Temperature dependence of effective interdiffusion coefficient of K^+ -for- Na^+ ($\bar{D}_{\text{Na-K}}$) for the boroaluminosilicate glasses with $x = 15$. The lines represent linear fits to the data using Eq. (2).

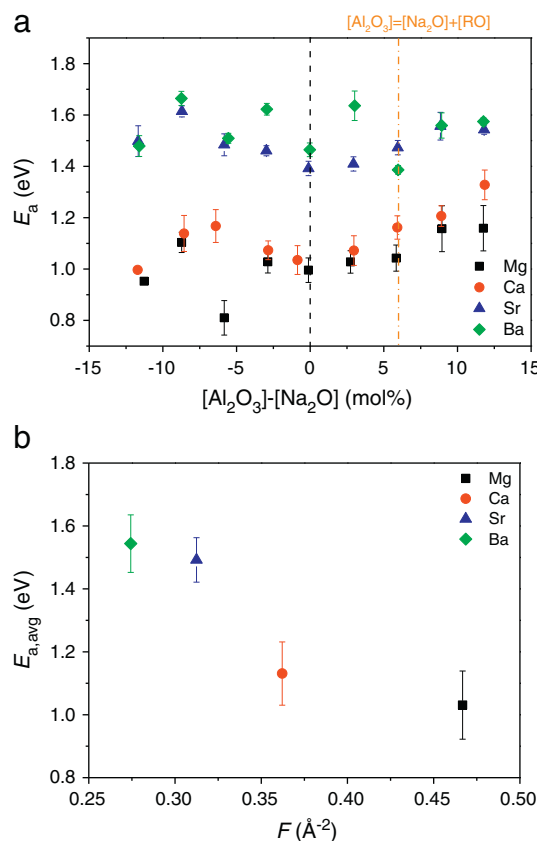


Fig. 5. (a) Composition dependence of activation energy (E_a) of K^+ -for- Na^+ interdiffusivity in the temperature range from 370 to 450 °C. (b) Average interdiffusion activation energy ($E_{a,\text{avg}}$) for glasses with varying $[\text{SiO}_2]/[\text{Al}_2\text{O}_3]$ ratio but fixed alkaline earth cation as a function of the divalent cation field strength (F). The field strength value, $F = Z/a^2$ where Z is ion charge and a is R–O bond distance, is calculated by assuming that all the divalent cations are in six-fold coordination [31].

4. Conclusions

The sodium–potassium interdiffusivity has been quantified at different temperatures in thirty-six alkaline earth sodium boroaluminosilicate glasses with varying alkaline earth modifier (Mg, Ca, Sr, Ba) and $\text{SiO}_2/\text{Al}_2\text{O}_3$ ratio. The results reveal a maximum of the alkali mobility at an intermediate $\text{SiO}_2/\text{Al}_2\text{O}_3$ ratio, suggesting that Na^+ diffuses faster when it acts as a charge compensator to Al^{3+} or B^{3+} than when it acts as a creator of non-bridging oxygen. The presence of relatively immobile alkaline earth cation in these glasses lowers the alkali diffusivity, an effect that is larger as the size of the divalent cation increases. However, in the peralkaline regime, the choice of alkaline earth oxide is less critical for optimizing interdiffusion, since the effect of the type of alkaline earth cation on the alkali diffusivity is much smaller than in the peralkaline regime. It is also found that there is a larger effect of increasing temperature on diffusivity for the larger alkaline earth cations than for the smaller ones. Results presented in this study are useful for optimizing basic compositions of boroaluminosilicate glasses in order to either reduce or enhance the alkali mobility.

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